Ryosei Tamura

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Education

Keio University April 2024 – Exp. March 2028
Science and Technology, Department of Information and Computer Science *Tokyo, Japan*

- **Relevant Coursework:** Machine Learning Fundamentals, Algorithms, Programming, Applied Mathematics, Computer Organization, Information Theory, Communication Systems, Electrical Circuits
- **Awards:** ServiceNow CreatorCon Hackathon 2025 Special Award

Skills

Programming Languages: Python (proficient), C (intermediate), JavaScript (intermediate)
ML Tools: PyTorch, pandas, NumPy, OpenCV, scikit-learn
Development Tools: Git, GitHub, VS Code
Language: Japanese (Native), English (Proficient, TOEFL iBT 106)

Research Experience

Carnegie Mellon University – Robotics Institute Summer Scholars (RISS) June 2026 – Present
Research Intern *Pittsburgh, PA*
Lab: Momentum Robotics Lab **Advisor:** Prof. Jeffrey Ichnowski

- Developed **SkillWeave**, a long-horizon dexterous manipulation framework that matches the demonstration method to each sub-task: teleoperation for reaching and transport, kinesthetic teaching for precise, contact-rich skills
- Designed an object-mask-conditioned diffusion policy that focuses on the manipulated object, not the demonstrator's hands in kinesthetic data; reached 65% dexterous sub-task success vs. 20% for prior SOTA
- Introduced successor-aware terminal steering, an inference-time method that guides each policy to end where the next one was trained to start; reached 27% end-to-end success on 3 real-world tasks vs. 5% without steering

Keio University December 2025 – Present
Student Researcher *Tokyo, Japan*
Lab: SMILAB **Advisor:** Prof. Komei Sugiura

- Analyzed misranking errors of Generalized Contrastive Learning (WWW 2025) for multi-modal retrieval, traced them to binary relevance bias, and proposed loss-weighting and evaluation fixes to use graded ranking signals
- Defined a new task, estimating each party's share of responsibility in ego-view accident videos; built an LLM-assisted labeling pipeline and benchmarked fine-tuned multimodal LLMs across three input types

Publications

- **Ryosei Tamura**, Xiaoxiang Dong, Uksang Yoo, Yuemin Mao, Romina Mir, Jonathan Francis, Jeffrey Ichnowski, "SkillWeave: Weaving Heterogeneous Demonstrations into Long-Horizon Manipulation Skills," arXiv:2610.12046, 2026 (under review)
- **Ryosei Tamura**, Xiaoxiang Dong, Uksang Yoo, Yuemin Mao, Romina Mir, Jonathan Francis, Jeffrey Ichnowski, "SkillWeave: Weaving Heterogeneous Demonstrations into Long-Horizon Manipulation Skills," *IROS 2026 Workshop on Enhancing Dexterous Manipulation Through Haptic Feedback* (poster)
- **Ryosei Tamura**, Andrew Shin, "Responsibility Distribution Estimation in Ego-View Accident Videos with Multimodal Large Language Models," *arXiv:2607.03591*, 2026

Relevant Experience

Oh my teeth, Inc. October 2025 – October 2026
AI Engineer/Leader (Intern) *Tokyo, Japan*

- Deployed a 2D CV method to quantify tooth rotation for orthodontic planning; adopted across 9 clinics
- Built a detection-to-segmentation pipeline: YOLO for per-tooth detection, SAM for masking, and contour-based geometry to estimate per-tooth orientation/rotation
- Implemented a Mask R-CNN baseline for per-tooth instance segmentation and tooth-ID assignment, improving accuracy via targeted augmentation, input-resolution tuning, and failure-case analysis

- Led a 5-person intern team and aligned definitions/metrics with clinicians to ensure outputs reflect clinical reasoning and expert insight

EXEO Digital Solutions, Inc. August 2025
Software Engineer (Intern) *Tokyo, Japan*

- Developed web and mobile applications for an access control system, covering the full development lifecycle from requirements definition and design to implementation

Volunteering & Leadership Experience

IEEE/RSJ IROS 2026 September 2026
Student Volunteer *Pittsburgh, PA*

Soccer Coach June 2024 – June 2026
Assistant Coach *Tokyo, Japan*

- Coached the Keio Junior High School soccer team; led the team to the Minato City tournament finals and qualified for the Tokyo Metropolitan tournament

Keio University Faculty of Science and Technology Festival April 2024 – October 2025
Performance Project Lead *Kanagawa, Japan*

- Led a 10-member team to plan and execute an a cappella performance, managing scheduling, coordination, and budget operations, resulting in a net profit of ¥25,500

Additional Experience

ServiceNow Japan Academic Program March 2025 – August 2025
Scholar *Tokyo, Japan*

- Built a job matching platform to connect employers and part-time job seekers using the ServiceNow platform
- Participated in ServiceNow CreatorCon Hackathon 2025, developing a hotel pre-check-in system to improve operational efficiency and deposit-based reservations; received a Special Award